AAE1001 - Introduction to Artificial Intelligence and Data Analytics in Aerospace and Aviation Engineering



Week 6 –
Github Destop and Test

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1 Create one repository for your group

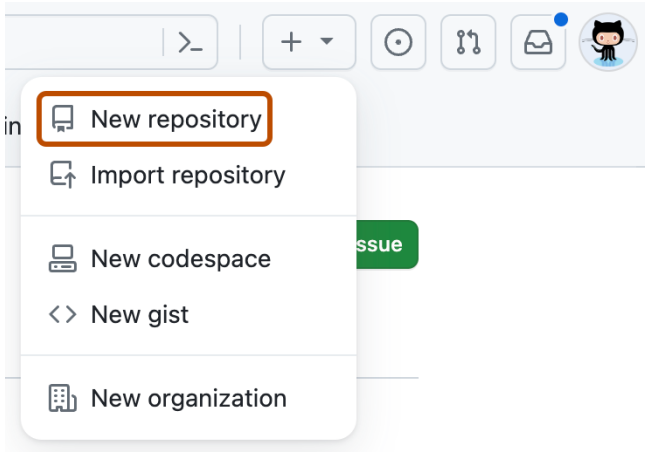
OWNER • GitHub website • One-time setup

1. Open github.com
Sign in. Click + → New repository.

2. Name your practice repository
Use aae1001-group-01-practice
(change 01 to your group number).

3. Choose Private + add README
A README creates the first commit.

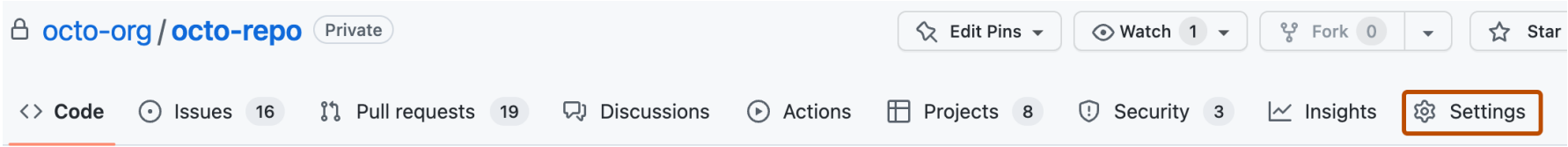
4. Click Create repository
Copy the repository URL for your group.



Ready first: GitHub account + GitHub Desktop + VS Code for every member.

2 Invite all your group members

OWNER + MEMBERS • GitHub website



1. **Owner: open Settings → Collaborators**
   Under Access, click Add people. Search each exact GitHub username.
2. **Owner: send each invitation**
   Click Add [username] to this repository. Repeat for everyone.
3. **Members: accept the invitation**
   Open the invitation link or email, accept, then open the shared repository URL.

Checkpoint: every member can open the same owner/repository on GitHub.

3 Clone the shared repository

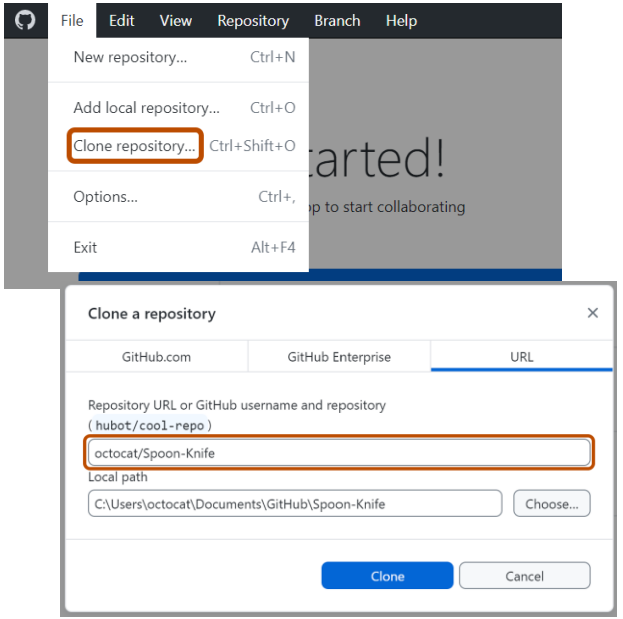
EVERYONE • GitHub Desktop

1. File → Clone repository
Choose the URL tab.

2. Paste the repository URL
Use your owner/repository,
not octocat/Spoon-Knife.

3. Choose a local folder
For example: Documents/GitHub.

4. Click Clone
Desktop creates your local copy.



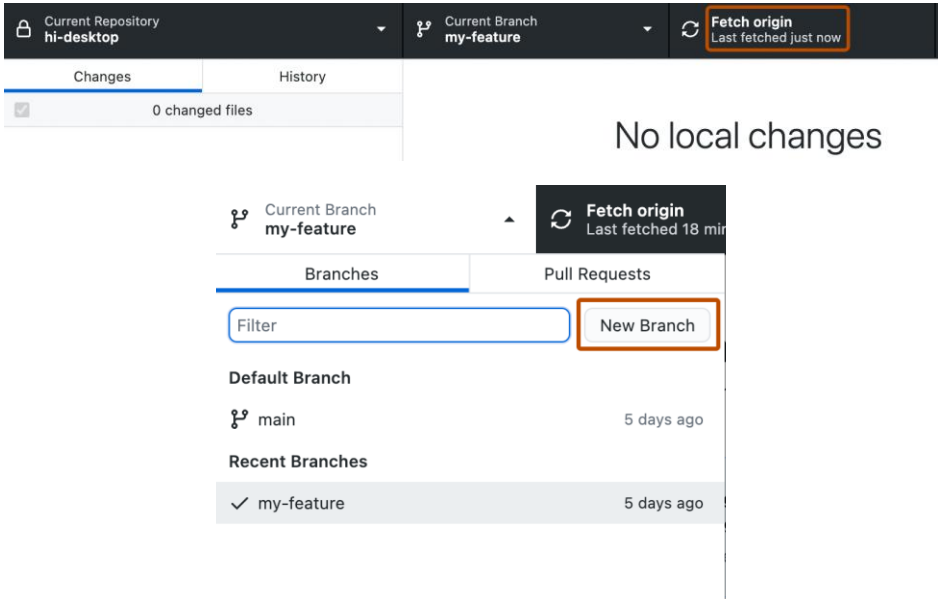
Checkpoint: Current repository shows your group's repository.

Screenshots: GitHub Docs • example names differ from our task

4 Update main, then create your branch

EVERYONE • GitHub Desktop • Repeat for each new contribution

1. **Select Current branch → main**
   Click Fetch origin; if shown, click Pull origin.
2. **Current branch → New Branch**
   Name it intro-YOUR-HANDLE.
3. **Base your branch on main**
   Click Create Branch.
4. **Check the branch label**
   It must show intro-YOUR-HANDLE before you start editing.



A branch is your separate line of work; main is the shared version.

Screenshots: GitHub Docs • example names differ from our task

5 Create one small Markdown file

EVERYONE • VS Code • Edit and save your local copy

1. Open the cloned folder in VS Code

Desktop: Repository →
Open in Visual Studio Code.

2. Create your Markdown file

Create members/YOUR-HANDLE.md
in the VS Code Explorer.

3. Write a short introduction

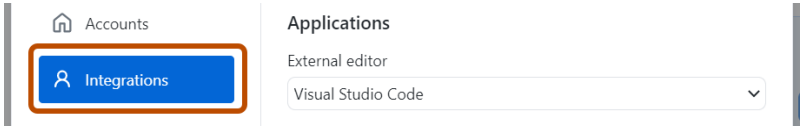
Use the example; replace every
YOUR-HANDLE with your username.

4. Save the file

Press Ctrl + S (Mac: Cmd + S).

```
# YOUR-HANDLE

- Interest: robotics
- Skill: Python beginner
- Goal: learn teamwork
```



File → Options → Integrations
Choose Visual Studio Code → Save

Checkpoint: the file is inside your cloned repository, in the members folder.

6 Review your change and commit

EVERYONE • GitHub Desktop • Save a local version

1. Return to Desktop → Changes

Click your file; inspect the change and keep its checkbox selected.

2. Write a short Summary

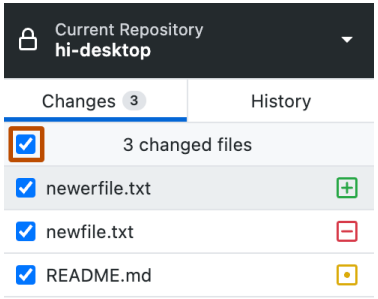
For example:
Add YOUR-HANDLE introduction

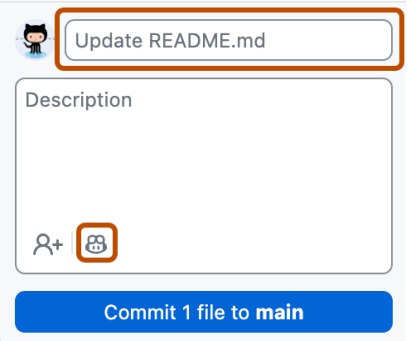
3. Check the destination branch

The commit button must name intro-YOUR-HANDLE.

4. Click Commit …

This records a version locally.
It has not been uploaded yet.





Example: main. Use your intro branch.

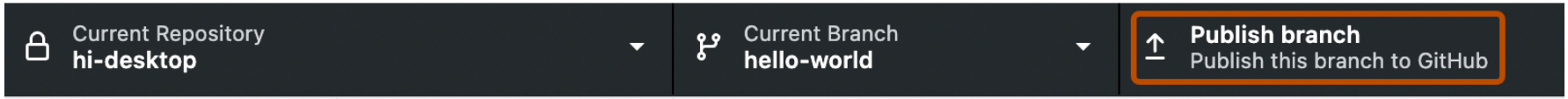
Checkpoint: the saved change appears in History on your branch.

Screenshots: GitHub Docs • example names differ from our task

7 Upload your branch to GitHub

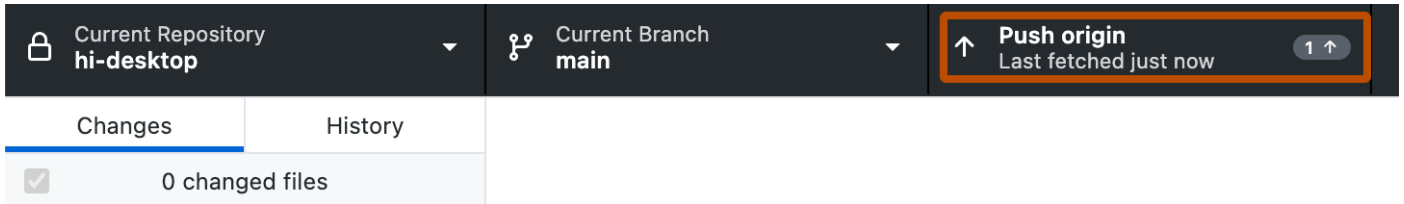
EVERYONE • GitHub Desktop • Local → remote

First upload: click Publish branch



Keep your intro-YOUR-HANDLE branch selected (screenshots use example names).

Later commits: click Push origin



Commit = save locally. Publish / Push = upload your branch. Main is unchanged.

Screenshots: GitHub Docs • example names differ from our task

8 Open a pull request for your change

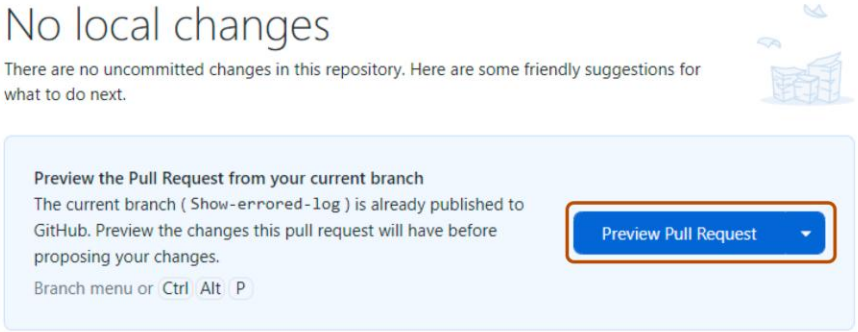
EVERYONE • Desktop → GitHub website

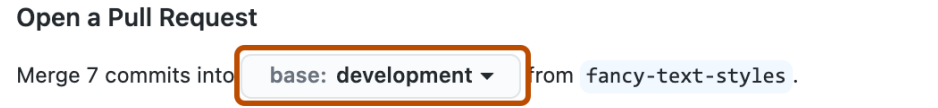
1. Click Preview Pull Request
Review the changes before sharing.

2. Check base: main
Compare / head: intro-YOUR-HANDLE.

3. Click Create Pull Request
A browser page opens.

4. Add a title and description
Say what changed; click
Create pull request on GitHub.



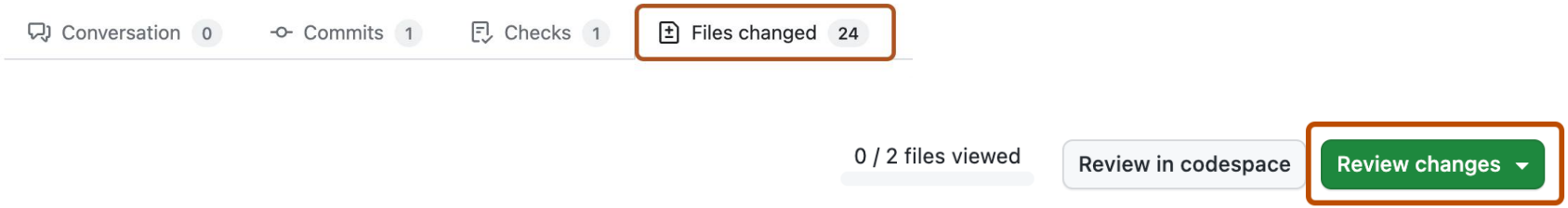


Set base to main; compare your intro branch.

Checkpoint: you have a PR link. Send it to the teammate reviewing your file.

9 Review a teammate's pull request

REVIEWER • GitHub website • A reviews B, B reviews C, …



1. Open the PR → Files changed

Read the actual file: correct path, username, useful introduction, no unrelated changes.

2. Click Review changes

Choose Approve if ready, or Request changes and explain the fix needed.

3. Click Submit review

Tell the coordinator when the PR is ready. You cannot approve your own PR.

If a fix is needed: author edits → saves → commits → pushes; the same PR updates.

Screenshots: GitHub Docs • example names differ from our task

10 Merge the reviewed contribution

COORDINATOR • GitHub website • One PR at a time

1. Check before merging

Base is main; teammate approved; only intended changes; no conflicts.

2. Click Merge pull request

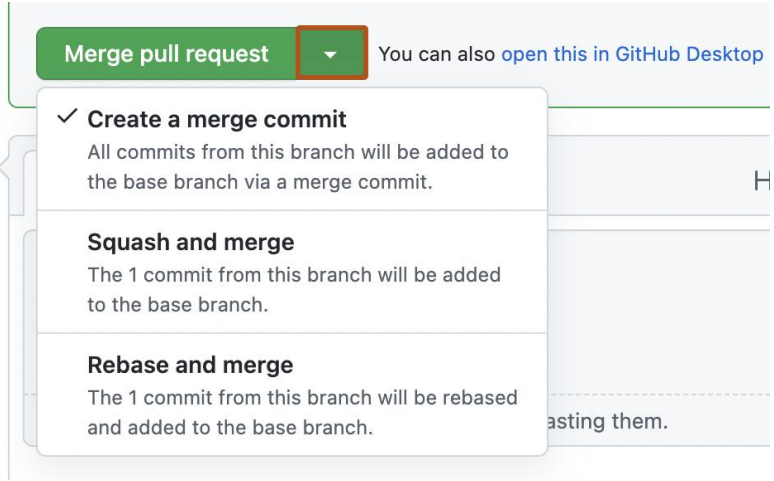
If a dropdown opens, choose Create a merge commit.

3. Click Confirm merge

GitHub adds the change to main.

4. Repeat for the other PRs

Each member's contribution should have its own reviewed PR.



Owner / coordinator needs write access.

Checkpoint: the PR shows Merged. Other computers still need to pull.

Screenshots: GitHub Docs • example names differ from our task

11 Bring the team's work to your computer

EVERYONE • GitHub Desktop • Remote → local

1. Select Current branch → main
Finish your own work before switching branches.

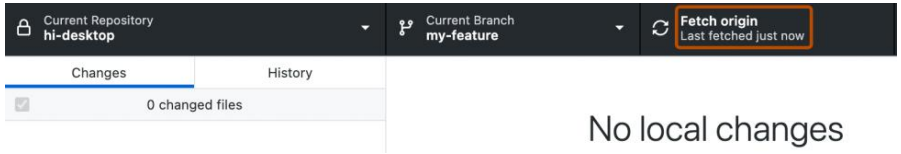
2. Click Fetch origin
Desktop checks GitHub for updates.

3. Click Pull origin, if shown
Download the merged changes.

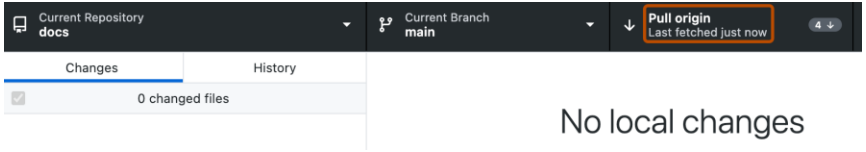
4. Open members in VS Code
Check that teammates' files are present in your local folder.



1 Switch back to main



2 Check for remote changes



For the next task: updated main → new branch → edit → commit → push → PR.

Screenshots: GitHub Docs • example names differ from our task

12 Mini-task: a contribution from everyone

TEAM CHALLENGE • About 25–30 minutes • Groups of 7–8

One group repository

Owner: create + invite.
Everyone: clone the same repository.

One contribution per student

Add members/YOUR-HANDLE.md.
Use your own branch and pull request.
Review the next person in your group.

Done means all four are true

1. One introduction file per member.
2. Each PR reviewed by a teammate.
3. All contributions merged into main.
4. Everyone pulled and can see them.

Show the TA: the members folder + reviewed, merged PRs + your updated local main.